JM0211
Process Mining

Explainable Predictive Process Monitoring

Does it matter *where* an activity happens? Apriori vs. IMPresseD itemsets for location importance

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1 Motivation and Scientific Background

Predictions need explanations

- **Outcome-oriented predictive process monitoring** predicts how a running case ends, so an owner can act in time [2].
- Black-box models (e.g. XGBoost) predict well but not *why* [2]; post-hoc explainable AI (XAI) explains a trained model [1].
- Existing methods say **which** activities matter, not **where in the trace**. **Location permutation importance** [4] moves a set of activities (an *itemset*) to other positions: if the model gets worse, their location matters.

> **Our question:** do other activities and locations come out as important if IMPresseD [5] chooses the itemsets instead of Apriori?

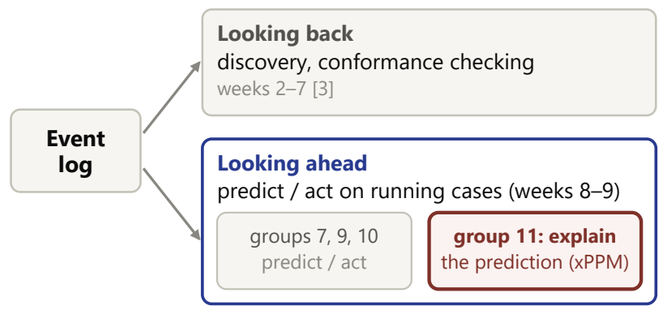
Use case: sepsis patients [4]

Predict whether a patient returns to the emergency room within 28 days. Six cases from [4, Table 1]:

1	RG **LA** LE CR ER ST IL IA NC ER CR LE CR	Negative
2	RG ER ST IL LE CR IA **LA** IC DI	Positive
3	RG **LA** ER ST LE CR IA NC CR LE CR LE CR	Negative
4	RG LE LE **LA** ER ST CR IA IC LE **LA** NC DI	Positive
5	RG ER ST LE CR IL IA **LA** NC CR LE	Positive
6	RG **LA** CR ER ST LE IL IA NC DI	Negative

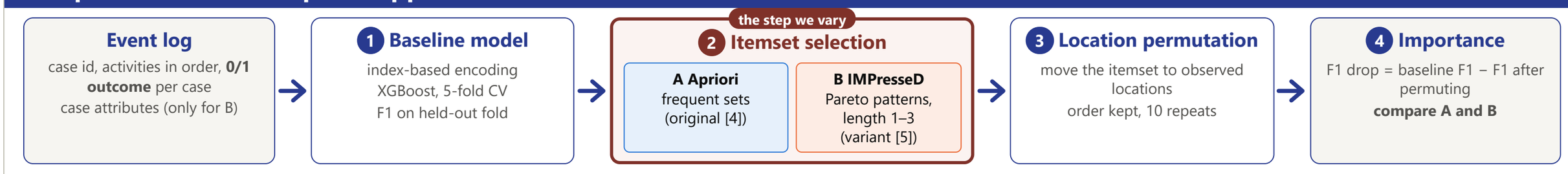
LA (lactic acid) is in **every** case: its *occurrence* explains nothing, its *location* does. At position 2 the outcome is **negative**, elsewhere positive.

Place in the course



Groups 9 and 10 change *how* a prediction is made, group 7 *acts* on it. We keep the model and **explain** what it learned from the activity order.

2 Explanation of Conceptual Approach



Input and output

- **In:** an event log with, per case, the **activities in execution order** and a **0/1 outcome**; control-flow only. BPIC11 has dates only, so events are ordered by **event number**.
- B also needs **case attributes** (BPIC11: age, diagnosis and three codes) for its case distance.
- **Out:** per itemset **50 F1 drops** (5 folds × 10 repeats), shown as mean and spread.

Parameters

Parameter	Value	Why
Min. support, top-k (A)	0.5, 10	defaults of [4]
Max. length (A, B)	3	assignment
Folds × repeats	5 × 10	as in [4]
Selection split (B)	80 / 20	B uses labels

Step 2: two ways to choose the itemsets

A · Apriori (original method)

- **Every single activity** + the **10 most frequent itemsets** of 2–3 activities (support ≥ 0.5).

Selects on: frequency; ignores the labels.

B · IMPresseD (our variant)

- Patterns: activities linked **directly** (a → b) or **eventually** (a ⇒ b), scored on three interests:

Outcome ↑	**Frequency** ↑	**Case distance** ↓
information gain on the label from the pattern count	share of cases with the pattern	cases with vs. without it (case attributes)

- Keep the **Pareto front**, extend it by one activity, repeat up to **length 3**. Then **drop the ordering**: a → b, b → a, a ⇒ b all become {a, b}.

Selects on: what the pattern count says about the outcome (80 % training split).

Step 3: location permutation

Index-based encoding: feature *i* = activity at position *i*, so the model sees **where** it happens.

	1	2	3	4	5
original	a	b	c	d	e
move **c**	**c**	a	b	d	e
move **{b, d}**	a	c	**b**	e	**d**

Only to positions observed for that activity in the log (**feasibility**); b before d stays b before d (**order**).

Implementation: Python port of the authors' code (pm4py, mlxtend, paretoset, XGBoost), one command `python main.py`, parameters in `config.yaml`. Fixes to the original: F1 drop on the held-out fold, every repeat starts from the original traces, itemset order always kept.

3 Evaluation Design and Results

Data

Log	Cases	Act.	Len.	Pos.	Use
BPIC11 f1	1130	164	36	40%	E3
BPIC11 f2	1130	207	40	78%	E3
BPIC11 f3	1111	156	31	23%	E3
Synthetic + 2 controls	15	8	7	60%	E1
Simulated, 2 × 10 logs	1000	16	22	52%	E2

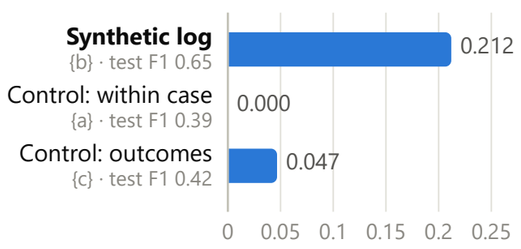
BPIC11: hospital logs of [4], labels of [2]. Len. = longest trace, Pos. = share of label 1.

Gold standard

- **Synthetic:** B is in every case; *early* B → label 0, *late* B → 1 (10 % noise). Two shuffled controls: nothing matters.
- **Simulated:** the location of **x** and of the block **p q** sets the outcome; only the *presence* of **y** matters; 10 noise activities.
- **BPIC11:** no ground truth, so we compare A and B.

E1 · Importance step

Does moving an activity whose location sets the label lower F1, while controls stay ≈ 0? Top itemset per log, mean F1 drop (A).



Moving **B** lowers F1 by **0.21** on average; in the controls no itemset moves it by more than ±0.08. The location is learnable (test F1 0.65 vs. 0.39 / 0.42). Only A tests B; IMPresseD does not select it. With 15 cases (3 test cases per fold) E1 is a sanity check.

E2 · Selection step

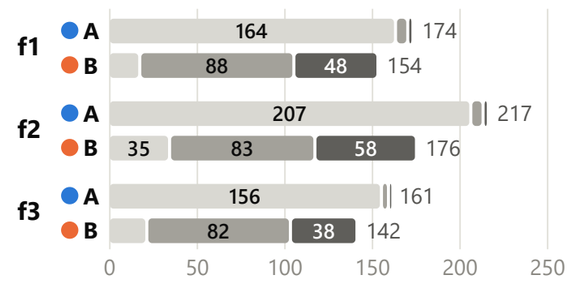
Is the planted itemset selected? (number of the 10 simulated logs)

Pair		{x} location	{p,q} location	{y} presence	Pure noise
Rare pair (30 %)	A	10/10	**0/10**	10/10	73%
	B	8/10	**0/10**	10/10	51%
Frequent pair (60 %)	A	10/10	**0/10**	10/10	73%
	B	10/10	**0/10**	10/10	49%

Both select the distractor {y} in every log; A always selects {x} (it keeps every activity), B misses it in 2 of 10 rare-pair logs. **No strategy ever selects {p, q}**: A ranks by frequency (its top 10 is full of start/end activities), and B only extends patterns whose *count* says something about the outcome, which p and q do not.

E3 · A vs. B on BPIC11: what is tested?

Selected itemsets by size: 1, 2, 3 activities

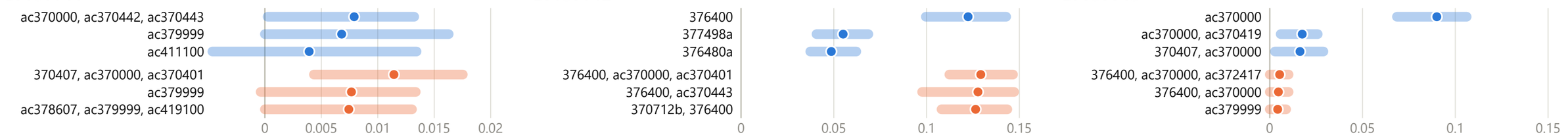


A tests mostly single activities, B mostly pairs and triples. Their multi-activity itemsets hardly overlap (Jaccard 0.00–0.02).

E4 · Sensitivity of A. Support 0.2 and top 25 add 15–20 multi-activity itemsets per log. The top itemset stays the same in f2 (376400, 0.12) and f3 (ac370000, 0.09); it changes in f1, where every drop is below 0.01.

E3 · Which locations matter? ● A Apriori ● B IMPresseD

Top 3 itemsets of each strategy per log: dot = mean F1 drop over 50 values, bar = 25th–75th percentile. Each log has its own scale.



f2: both point to **376400**. A tests it alone (0.122); B's 24 best itemsets all contain it (best 0.129, alone 0.121), so the added activities change little. **f3:** A's top is **ac370000** (0.090); B does not select it as a single activity; its best itemset reaches only 0.005. **f1** (label on occurrence only): every mean drop ≤ 0.011, location hardly matters.

4 Limitations and Future Work

Limitations

- **Selection ≠ importance.** Both strategies select on *existence* (frequency, pattern count); the importance step tests *location*. A pair whose location matters but whose presence does not is never tested (E2).
- An itemset inherits the importance of its strongest activity (f2: every top itemset of B contains 376400).
- B selects on 80 % of the cases, but the folds use all cases. One model, one hospital, date-only timestamps, a tiny synthetic log.

Future work

- **Location-aware selection:** an IMPresseD interest on the *position* of a pattern instead of its count.
- Report the gain of an itemset over its best subset; repeat the selection inside each fold.
- Other models and encodings (e.g. LSTM, position from the end) and domains; permute a pattern as a whole, with its order.

Take-away

The importance step finds a planted location (E1), but only what is selected can be found.

A and B test different itemsets; on BPIC11 the strongest location effects are single activities, which A always tests and B can miss (f3).

1. A. B. Arrieta et al. Explainable artificial intelligence (XAI). *Information Fusion* 58, 2020.
2. I. Teinemaa, M. Dumas, M. La Rosa, F. M. Maggi. Outcome-oriented predictive process monitoring: review and benchmark. *ACM TKDD* 13(2), 2019.
3. W. M. P. van der Aalst. *Process Mining: Data Science in Action*, 2nd ed. Springer, 2016.
4. M. Vazifehdoostirani, M. Abbaspour Onari, I. Grau, L. Genga, R. Dijkman. Uncovering the hidden significance of activities location in predictive process monitoring. *ICPM 2023 Workshops*, LNBIP 503, 2024.
5. M. Vazifehdoostirani, L. Genga, X. Lu, R. Verhoeven, H. van Laarhoven, R. Dijkman. Interactive multi-interest process pattern discovery. *BPM 2023*, LNCS 14159, 2023.

Code: github.com/HoyngM/group-11-pm